

Automotive Parts Assistant

An AI-powered assistant for automotive parts search, stock checking, ordering, and demand forecasting.

Features

- RAG-based product retrieval
- LLM decision-making (Ollama - LLaMA 3)
- Tool calling:
 - `check_stock`
 - `create_order`
 - `find_parts_by_vehicle`
- Chat-style CLI interface
- Guardrails (off-topic protection)
- Clarification handling
- Demand Forecasting (Part B)

Project Structure

```
root/
├── assistant/
│   ├── agent.py
│   ├── tools.py
│   └── rag.py
├── data/
│   ├── catalogue.csv
│   └── sales.csv
├── forecasting/
│   └── forecast.py
├── DESIGN.md
└── README.md
```

Setup

1. Install dependencies

```
pip install pandas numpy ollama
```

2. Install Ollama

Download from:

<https://ollama.com>

Then run:

```
ollama pull llama3
```

Run Assistant

```
python assistant/agent.py
```

Example Queries

Product Query

Do you have brake pads?

Vehicle Query

Show parts for Meteor 350

Order Query

I want to buy oil filter

Guardrails

Rejects off-topic queries:

What is the weather?

→ Not supported

Clarification

Handles ambiguity:

show parts

→ asks clarification

Forecasting (Part B)

Run:

```
python forecasting/forecast.py
```

Output:

- Per-SKU performance
- Baseline vs Model comparison
- MAE & MAPE metrics

Model Performance

- Baseline MAE: ~9.35
- Model MAE: ~8.29
- Improvement achieved ✓

Evaluation

Tested across:

- Guardrails
- Ambiguity handling
- Tool accuracy
- Grounded responses

Future Improvements

- Vector search (FAISS)
- UI (WhatsApp style)
- Multimodal input (image → part detection)
- Advanced forecasting

Author

AI/ML Internship Assignment Submission

✓ Status

- ✓ Fully functional
- ✓ Meets all requirements
- ✓ Ready for evaluation